

LukeDoesMaths

LEAVING CERTIFICATE NOTES

MATHEMATICS --- HIGHER LEVEL

Differentiation

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1 What is a Derivative?

The derivative of a function tells us its *rate of change* at a point — geometrically, the slope of the tangent to the curve at that point.

DEFINITION

For a function $f(x)$, the derivative $f'(x)$ is defined as

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

This is sometimes called *differentiation from first principles* and is examinable at Higher Level.

EXAM TIP

First-principles questions almost always appear as a stand-alone Paper 1 question. Learn the definition **exactly** as written — markers look for the correct limit notation, not just the final answer.

2 Differentiation Rules

Power rule: $\frac{d}{dx}(x^n) = nx^{n-1}$

Product rule: $(uv)' = u'v + uv'$

Quotient rule: $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$

Chain rule: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

WORKED EXAMPLE

Differentiate $f(x) = 3x^4 - 5x^2 + 2x - 7$ with respect to x .

Solution. Apply the power rule to each term:

$$f'(x) = 12x^3 - 10x + 2.$$

WORKED EXAMPLE – CHAIN RULE

Differentiate $g(x) = (3x^2 + 1)^5$.

Solution. Let $u = 3x^2 + 1$, so $g = u^5$.

$$\frac{dg}{du} = 5u^4, \quad \frac{du}{dx} = 6x$$

$$\frac{dg}{dx} = 5u^4 \cdot 6x = 30x(3x^2 + 1)^4.$$

2.1 Common Mistakes

- Forgetting to multiply by the derivative of the inside function (chain rule).
- Applying the product rule to a single term multiplied by a constant — constants just carry through.
- Dropping the negative sign when differentiating $\frac{1}{x^n} = x^{-n}$.

3 Applications

3.1 Applied Maths link: Velocity and Acceleration

DEFINITION – KINEMATICS

If $s(t)$ is displacement as a function of time,

$$v(t) = \frac{ds}{dt}, \quad a(t) = \frac{dv}{dt} = \frac{d^2s}{dt^2}.$$

WORKED EXAMPLE – APPLIED MATHS

A particle moves so that $s(t) = t^3 - 6t^2 + 9t$ (in m, t in s). Find the particle's velocity and acceleration at $t = 2$ s.

Solution.

$$v(t) = 3t^2 - 12t + 9 \Rightarrow v(2) = 12 - 24 + 9 = -3 \text{ m/s}$$

$$a(t) = 6t - 12 \Rightarrow a(2) = 0 \text{ m/s}^2$$

3.2 Physics link: Instantaneous Rate of Change

EXAM TIP

In Physics, differentiation underpins the definitions of instantaneous velocity and instantaneous acceleration, and appears again in topics like induced EMF ($\mathcal{E} = -\frac{d\Phi}{dt}$). Recognising the same mathematical structure across subjects saves study time.

4 Summary

- The derivative gives the instantaneous rate of change / gradient of the tangent.
- Know the power, product, quotient and chain rules fluently.
- First-principles differentiation must use the limit definition exactly.
- Differentiation connects directly to kinematics (Applied Maths) and rates of change in Physics.